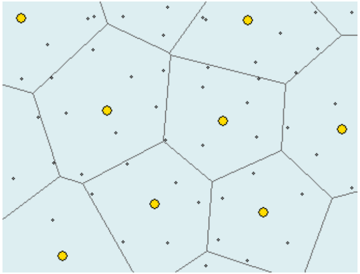
K-means: nearest centres and Voronoi regions

Choose K, then repeat

1. Assign every cell to its nearest centre.
2. Update each centre to its cluster mean.

K: number of clusters. Each cell gets one label.

Geometric model
Convex regions with linear boundaries.
Probabilistic model: hard-assignment Gaussian clusters with a shared spherical covariance and equal prior weights.



Yellow points: centres. Each region contains points closest to its centre.

Our data: 20,000 live PBMCs, 37 markers, 20 donors.

Which choice of K resolves meaningful populations, including potentially rare subsets?

Figure excerpt: Ulrike von Luxburg, Statistical Machine Learning, Summer 2024, slide 840 (PDF p. 844), cropped. Gaussian interpretation: Macke / Hennig, Lecture 11.

Silhouette and Davies–Bouldin: what gets averaged?

Silhouette: cell-weighted average

$$\bar{s} = \sum_{c=1}^{K} \frac{n_c}{N}\,\bar{s}_c$$

n_c: number of cells in cluster c.
N: total number of cells.
$\bar{s}_c$: mean silhouette in cluster c.

A group with **1 % of cells** has **1 % weight** for its own mean silhouette.

DB: equal weight per cluster

$$DB = \frac{1}{K}\sum_{c=1}^{K} \max_{j \neq c} \frac{S_c + S_j}{\lVert \mu_c - \mu_j \rVert}$$

S_c: mean distance of cells to their centre.
μ_c: centre (mean vector) of cluster c.

Each cluster has $1/K$ **weight**. Lower is better.

DB gives every cluster same weight.

Definitions: scikit-learn clustering documentation.

Subsampling within the observed cells

Why keep the initial sample?

- A fresh sample may omit a rare population that we observed initially.
- Different cells also prevent a direct membership comparison.
- We first test how well observed groups survive removing cells.

Our stability experiment

1. Keep 80 % of cells within each donor, without replacement.
2. Refit scaling and clustering with the same parameter settings.
3. Compare both partitions on the **same retained cell IDs**.

Ten repeats, 16,000 cells each, with changing random seeds.

This measures stability conditional on our initial sample. Even 80 % subsampling can lose a tiny group.

Our notebook implementation.

Stability: global ARI or cluster-wise Jaccard?

ARI compares entire partitions

It measures agreement over cell pairs and adjusts for chance.

A group of n cells contains $n(n-1)/2$ within-group pairs. Large groups therefore contribute far more pairs.

Splitting a rare group changes relatively few pairs, so global ARI can remain high.

Jaccard follows each population

$$J(C, D) = \frac{|C \cap D|}{|C \cup D|}$$

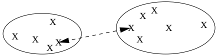
C: reference cluster among the retained cells.
D: cluster from the new fit on those same cells.

Our choice: Jaccard per cluster, then equal cluster weights. Both approaches work without ground truth.

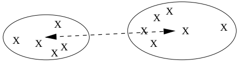
SOURCE

Agglomerative clustering: which groups merge?

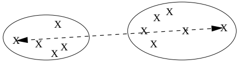
Single linkage: $dist(C, C') = \min_{x \in C, y \in C'} d(x, y)$



Average linkage: $dist(C, C') = \frac{\sum_{x \in C, y \in C'} d(x,y)}{|C| \cdot |C'|}$



Complete linkage: $dist(C, C') = \max_{x \in C, y \in C'} d(x, y)$



C, C': two clusters; x, y: cells.
$d(x, y)$: Euclidean distance; $|C|$: cell count.

Start with one cluster per cell
Merge the closest groups repeatedly. Cut the resulting tree at the desired cluster count.

Ward uses a different merge rule
Choose the smallest increase in within-cluster squared error (very k-means like). This favours compact groups.

Earlier merges remain fixed in every variant.

Figure excerpt: Ulrike von Luxburg, Statistical Machine Learning, Summer 2024, slide 850 (PDF p. 854), cropped. Ward: SciPy linkage documentation.

Leiden 1/2: the graph and the objective

Build a graph from marker profiles

- Each cell becomes a node.
- Find neighbouring cells in the 37-dimensional marker space.
- Weighted edges represent local similarity between cells.

Neighbour count controls the scale of the graph. Our selected graph uses 30 neighbours.

Find communities in that graph

Leiden searches for a partition with a high graph-quality score. HIER DIE BEIDEN

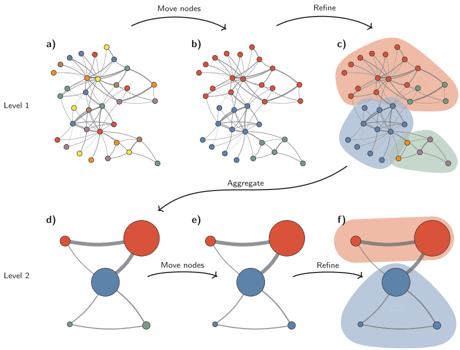
In our setup, the score rewards more internal connection weight than expected from node connectivity.

Resolution controls the penalty for the expected connections. Higher values usually yield smaller communities.

We set graph and resolution parameters; the number of clusters emerges. Here: resolution 1.2, 18 groups.

Traag, Waltman & van Eck (2019), pp. 1–2; our notebook uses Scanpy / leidenalg. Graph construction precedes Leiden. The selected objective is the configuration-model (RB) objective, not CPM.

Leiden 2/2: move, refine and aggregate



1. Move nodes (a–b)
Start with one group per node. Move nodes when this improves the quality score.

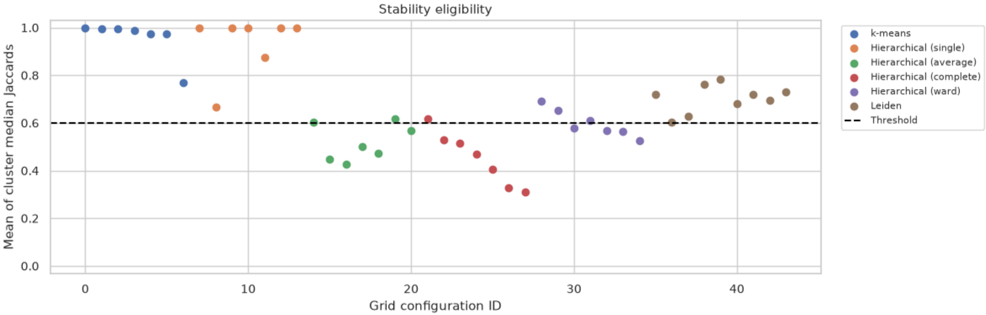
2. Refine groups (b–c)
Within each group, rebuild connected subgroups. Weakly joined parts can stay separate.

3. Aggregate (c–d)
Each refined subgroup becomes a node. Repeat on the smaller graph (d–f).

Refinement keeps subgroups available for later moves, instead of permanently collapsing an entire group. Improvement from Louvain.

Figure 3, cropped: Traag, Waltman & van Eck (2019), From Louvain to Leiden, CC BY 4.0.

Selection: stability first, then Davies–Bouldin



Each dot: one configuration. Keep mean stability ≥ 0.6 (arbitrary), then minimise DB per method.

K-means++ and 10 starts
Reduce sensitivity to poor initial centres.
Stochastic **farthest first** heuristic

Single linkage and chaining
A large group can grow through large chain of local neighbourhoods

Original saved notebook plots.

Excellent scores can describe an unhelpful partition

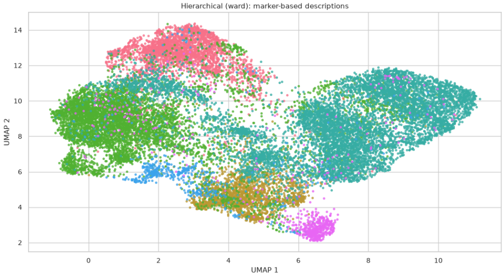
Method	Parameters	Clusters	DB ↓	Stability	Largest cluster
K-means	$k = 4$	4	2.363	0.995	37.780 %
Single	$k = 2$	2	0.264	1.000	99.995 %
Average	$k = 2$	2	0.894	0.602	99.975 %
Complete	$k = 2$	2	0.948	0.617	99.960 %
Ward	$k = 6$	6	2.522	0.611	38.220 %
Leiden	$n = 30, r = 1.2$	18	2.361	0.683	17.940 %

Single linkage: 19,999 cells + 1 cell
DB = 0.264 and mean stability = 1.

An isolated event can score well and recur. These metrics alone do not establish a rare cell population.

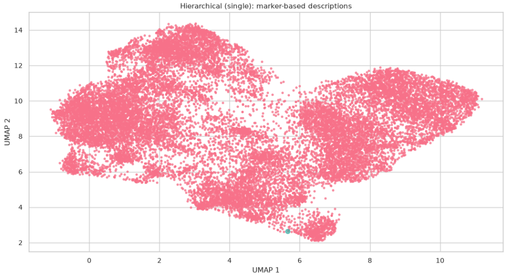
Ward and single linkage on the same UMAP

Ward: 6 clusters



DB: **2.522** Mean stability: **0.611**

Single: 19,999 cells + 1 cell



DB: **0.264** Mean stability: **1.000**

Ward divides the broad cloud. Single linkage assigns almost the entire cloud to one cluster.

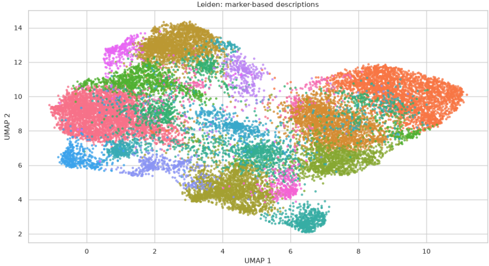
Original saved notebook plots.

K-means and Leiden give different resolutions

K-means: 4 clusters

3,148–7,556 cells per cluster
DB: **2.363** Mean stability: **0.995**

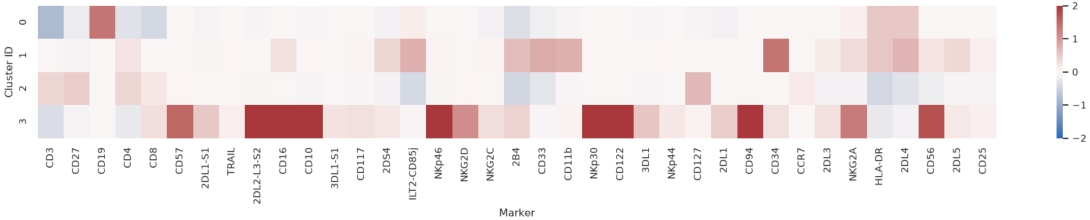
Leiden: 18 clusters



223–3,588 cells per cluster
DB: **2.361** Mean stability: **0.683**

Original notebook plots on the same UMAP coordinates.

K-means: marker profiles of the four groups



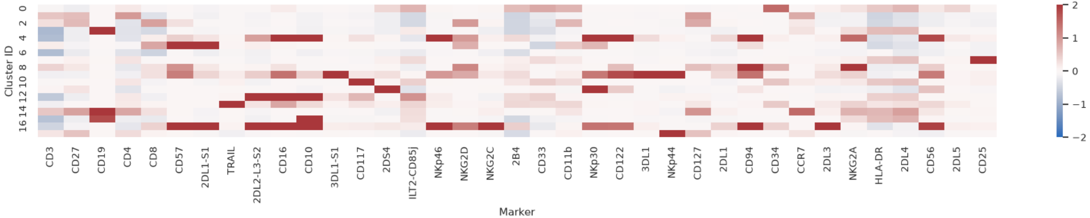
$$\text{Relative marker score} = \frac{\text{cluster median} - \text{sample median}}{\text{sample IQR}} \text{ vs. } \frac{\text{cluster mean} - \text{sample mean}}{\text{sample standard deviation}}$$

For each marker after arcsinh: $\mathrm{IQR} = Q_{75\%} - Q_{25\%}$. Red: higher median; blue: lower.

more robust statistical marker

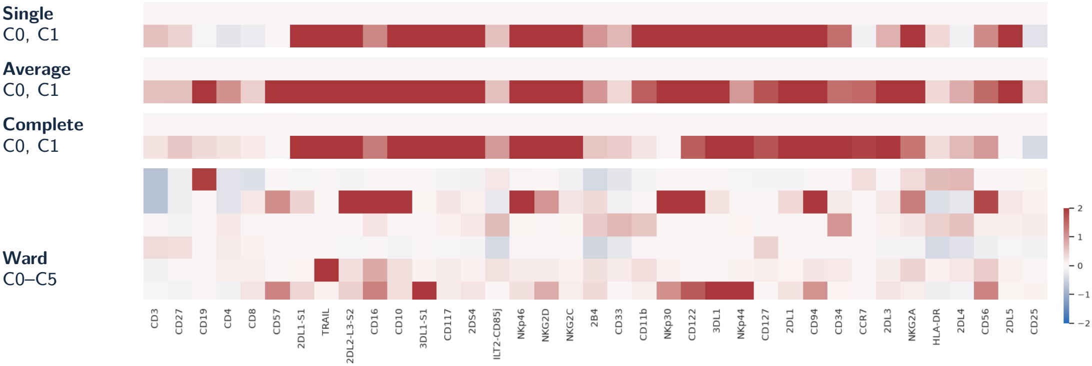
Original notebook panel. NIST

Leiden: marker profiles of the finer groups



Original notebook panel.

Hierarchical clustering: marker profiles



Single / average / complete: tiny groups with broadly elevated markers. Ward: more differentiated profiles.

Original notebook heatmap strips.

Assessment for the cell-population question

K-means: broad, reproducible groups
4 clusters, mean stability 0.995.
Broad B-cell-like and NK-cell-like marker profiles.

Ward: intermediate resolution
6 clusters with differentiated marker profiles.
Mean stability 0.611; 3 of 6 groups below 0.6.

Leiden: finer groups
18 clusters, mean stability 0.683.
More detail, but 4 of 18 groups below 0.6.

Single, average and complete
Selected solutions mainly separate a few events from almost all other cells.

Rare cell populations
Equal cluster weighting helps evaluate small groups. A biological interpretation also needs coherent markers and replication.

K-means provides a stable baseline. Ward and Leiden add detail with weaker reproducibility.

Our saved results for 20,000 live PBMCs.

Backup: K-means initialisation and Ward merges

K-means++ and repeated starts

- Standard k-means++ samples centres in proportion to squared distance from existing centres.
- scikit-learn uses a greedy variant with several candidates at each step.
- Our `n_init=10` retains the run with the lowest inertia.
- A fixed seed reproduces a run. Our resampling also changes the seed.

Ward minimises the increase in SSE

$$\Delta(A, B) = \frac{n_A n_B}{n_A + n_B} \|\mu_A - \mu_B\|^2$$

Ward prefers merges that add little within-cluster variation.

n_A, n_B: numbers of cells in groups A, B. μ_A, μ_B: their mean vectors. Δ: added squared error.

Ward only merges. K-means can reassign cells.

scikit-learn KMeans and SciPy linkage documentation. SSE = sum of squared errors.

Backup: geometry and the singleton example

Silhouette

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

$a(i)$: mean distance to other cells in its cluster.
$b(i)$: smallest mean distance to another cluster.

Resolving a nearby subgroup can reduce silhouette even if the subgroup is biologically meaningful. This is a possibility, not an observed comparison in our run.

Davies–Bouldin with cluster and singleton

$$S_c = \frac{1}{n_c} \sum_{x_i \in C_c} \|x_i - \mu_c\|$$

$$DB = \frac{S_{\text{large}} + 0}{\|\mu_{\text{large}} - x_{\text{singleton}}\|}$$

C_c: cluster c; n_c: its cell count. x_i: a cell vector; μ_c: the cluster mean.
A distant singleton has zero scatter and can yield a low DB.

scikit-learn silhouette and Davies–Bouldin definitions. No claim that silhouette always selects too few clusters, or that DB is neutral across cluster shapes.

Backup: sensitivity to the stability cutoff

Mean stability cutoff	0.50	0.60	0.65	0.70	0.75
K-means: selected clusters	4	4	4	4	4
Leiden: selected clusters	18	18	18	7	6
Ward: selected clusters	6	6	2	—	—

The cutoff affects the chosen resolution.

K-means remains at four clusters over these tested thresholds.

Stricter thresholds select coarser Leiden solutions.

"—" means no eligible configuration. Single linkage remains at $k = 2$ throughout.

The 0.6 threshold is a transparent heuristic. Sensitivity analysis shows which conclusions depend on it.

Backup: from K-means loss to Gaussian likelihood

$$
\begin{aligned}
(r, m) &= \underset{r,m}{\arg\min} \sum_{i=1}^{N} \sum_{k=1}^{K} r_{ik} \lVert x_i - m_k \rVert^2 \\
&= \underset{r,m}{\arg\max} \left[\sum_{i=1}^{N} \sum_{k=1}^{K} r_{ik} \left(-\frac{\lVert x_i - m_k \rVert^2}{2\sigma^2} \right) + \text{const.} \right] \\
&= \underset{r,m}{\arg\max} \prod_{i=1}^{N} \sum_{k=1}^{K} r_{ik} \frac{\exp\left(-\lVert x_i - m_k \rVert^2 / (2\sigma^2)\right)}{Z} \\
&= \underset{r,m}{\arg\max} \prod_{i=1}^{N} \sum_{k=1}^{K} r_{ik} \mathcal{N}(x_i; m_k, \sigma^2 I_d) \\
&= \underset{r,m}{\arg\max}\, p(X \mid m, r).
\end{aligned}
$$

Adapted from Macke / Hennig, Probabilistic Machine Learning (2024), Lecture 11, slide 15, . Binary one-hot r; fixed shared $\sigma^2 > 0$; $Z = (2\pi\sigma^2)^{d/2}$. Component index corrected to m_k.

Sources

Analysis and lecture material

`notebooks/03_clustering.ipynb`, run on 7 September 2026.

$k \in \{2, 3, 4, 6, 8, 10, 12\}$; neighbours $\in \{10, 30, 60\}$; resolutions $\in \{0.3, 0.7, 1.2\}$.

Initial seed 42, 10 K-means starts, 44 reference fits and 440 subsampling fits.

Ulrike von Luxburg, *Statistical Machine Learning*, Summer 2024, slides 840 and 850 (cropped figures).

Jakob H. Macke / original contributions P. Hennig, *Probabilistic Machine Learning*, Summer 2024, Lecture 11, slide 15. Adapted derivation under CC BY-NC-SA 3.0.

National Institute of Standards and Technology NIST

Biology and graph clustering

Arvaniti and Claassen (2017). Sensitive detection of rare disease-associated cell subsets.

Traag et al. (2019). From Louvain to Leiden: guaranteeing well-connected communities.

Definitions and implementation

scikit-learn: clustering metrics KMeans SciPy: linkage Scanpy: Leiden